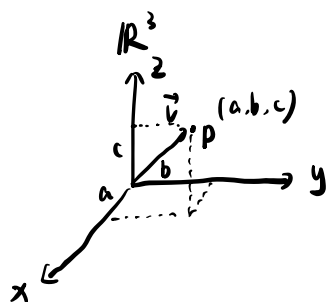
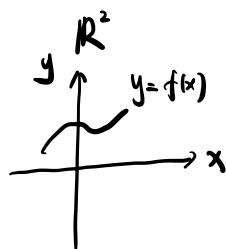


Lecture 2 Three - Dimensional Space

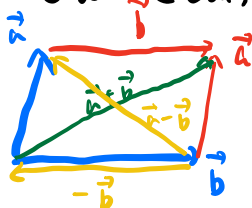


$$\vec{v} = a\vec{i} + b\vec{j} + c\vec{k} = \langle a, b, c \rangle$$

$$\vec{a} = \langle a_1, a_2, a_3 \rangle, \quad \vec{b} = \langle b_1, b_2, b_3 \rangle$$

$$\text{Vector addition: } \vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

$$\text{Scalar multiplication } c\vec{a} = \langle ca_1, ca_2, ca_3 \rangle$$



$$\text{Properties. } \vec{a} + \vec{b} = \vec{b} + \vec{a}$$

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$$

$$(\lambda + \mu)\vec{a} = \lambda\vec{a} + \mu\vec{a}, \quad \lambda(\vec{a} + \vec{b}) = \lambda\vec{a} + \lambda\vec{b}$$

Dot Product.

Def. Let $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$. then dot product

$$\vec{a} \cdot \vec{b} \triangleq a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\vec{a} \cdot \vec{a} = a_1^2 + a_2^2 + a_3^2$$

$$\text{The length of } \vec{a} \text{ is } |\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\text{Properties. } \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

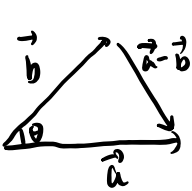
$$(\vec{a} + \vec{b}) \cdot \vec{c} = \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}$$

$$(\lambda \vec{a}) \cdot \vec{b} = \lambda \vec{a} \cdot \vec{b}$$

$$\vec{0} \cdot \vec{a} = \vec{a} \cdot \vec{0} = 0$$

Theorem. $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$, θ is the angle between $\vec{a}$ and $\vec{b}$,

$$\text{Then } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

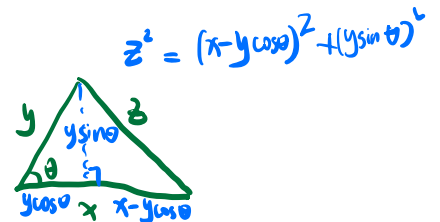


$$\text{Pf. } |\vec{a}-\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos\theta$$

$$(\vec{a}-\vec{b}) \cdot (\vec{a}-\vec{b}) = \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} - 2|\vec{a}||\vec{b}|\cos\theta$$

$$-2\vec{a} \cdot \vec{b} = -2|\vec{a}||\vec{b}|\cos\theta$$

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$$



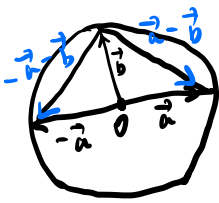
$$z^2 = x^2 + y^2 - 2xy \cos \theta$$

Law of Cosines

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} \right)$$

$\vec{a} \perp \vec{b}$ if and only if $\vec{a} \cdot \vec{b} = 0$, for non-zero $\vec{a}$ and $\vec{b}$.

Example. Show that any triangle which is inscribed in a circle and has one side coincide the diameter of the circle must be a right-angled triangle.



$$\text{Pf. } (-\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})$$

$$= -\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b}$$

$$= -|\vec{a}|^2 + |\vec{b}|^2$$

$$= 0$$

Cross Product.

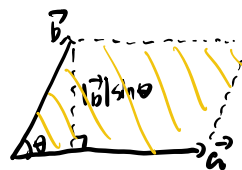
Def. $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$, θ is the angle between $\vec{a}$ and $\vec{b}$.

then cross product $\vec{a} \times \vec{b}$ is a **vector** defined as.

• the length is $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$

• $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$

• direction of $\vec{a} \times \vec{b}$ is determined by the right-hand grab rule.

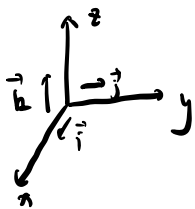


Properties. 1) $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

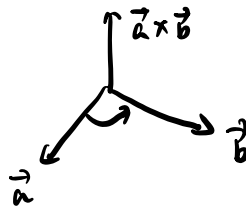
2) $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$

3) $\vec{a} \times \vec{0} = \vec{0}$

4) $\vec{a} \times \vec{a} = \vec{0}$



$$\vec{i} \times \vec{j} = \vec{k}, \quad \vec{j} \times \vec{k} = \vec{i}, \quad \vec{k} \times \vec{i} = \vec{j}$$



Theorem. $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

Determinant of a matrix $= (a_2 b_3 - a_3 b_2) \vec{i} - (a_1 b_3 - a_3 b_1) \vec{j} + (a_1 b_2 - a_2 b_1) \vec{k}$.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Pf. $\vec{a} \times \vec{b} = (a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}) \times (b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k})$
 $= \underline{a_1 b_2 \vec{k}} - \underline{a_1 b_3 \vec{j}} - \underline{a_2 b_1 \vec{k}} + \underline{a_2 b_3 \vec{i}} + \underline{a_3 b_1 \vec{j}} - \underline{a_3 b_2 \vec{i}}$

□

Example. A(0, 2, -1) B(4, 0, -1) C(7, -3, 0)

1) Find a vector $\vec{n}$ which is orthogonal to the plane passing through A, B and C

2) Find the area of $\triangle ABC$.

Sol. $\vec{AB} = \langle 4, -2, 0 \rangle$

$\vec{AB} \times \vec{AC} = \langle -2, -4, -6 \rangle$

$\vec{AC} = \langle 7, -5, 1 \rangle$

$\vec{n} = \langle 1, 2, 3 \rangle$

Area of $\triangle ABC = \frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2} \sqrt{(-2)^2 + (-4)^2 + (-6)^2} = \sqrt{14}$.